



Chronic exposure to polyvinyl chloride microplastics induces liver injury and gut microbiota dysbiosis based on the integration of liver transcriptome profiles and full-length 16S rRNA sequencing data

Xuebing Chen^{a,b,1}, Jingshen Zhuang^{c,1}, Qianling Chen^{a,b}, Luyao Xu^{a,b}, Xia Yue^{a,b}, Dongfang Qiao^{a,b,*}

^a Guangzhou Key Laboratory of Forensic Multi-Omics for Precision Identification, School of Forensic Medicine, Southern Medical University, Guangzhou, Guangdong 510515, China

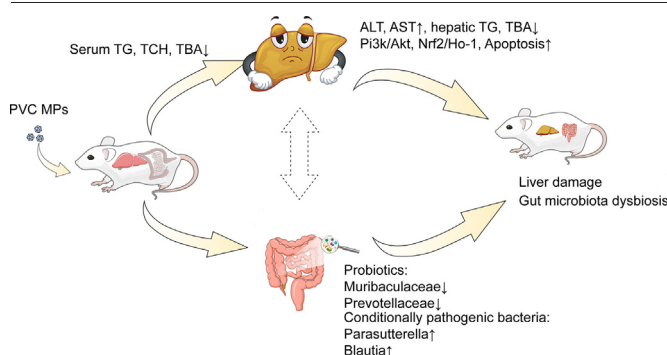
^b School of Forensic Medicine, Southern Medical University, Guangzhou 510515, China

^c Division of Spine Surgery, Department of Orthopaedics, Nanfang Hospital, Southern Medical University, Guangzhou 510515, China

HIGHLIGHTS

- Adult mice exposure to PVC-MPs (2 μ m, 0.5 mg/day) for 60 days induced hepatotoxicity.
- More than 1500 DEGs were identified by RNA-seq of the livers of PVC-MPs-treated mice.
- PVC-MPs induced PI3K/Akt-related apoptosis and oxidative stress in mouse livers.
- PVC-MPs exposure changed the structure of the gut microbiota in mice.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Jay Gan

Keywords:

Microplastics
Polyvinyl chloride
Hepatotoxicity
Gut microbiota
PI3K/Akt

ABSTRACT

Microplastics (MPs) have become harmful environmental pollutants, and their potential toxicity to organisms has attracted extensive attention. However, the effects of polyvinyl chloride MPs (PVC-MPs) on the liver and their associated mechanism in mice remain obscure. Here, male mice were exposed to 2- μ m PVC-MPs (0.5 mg/day) for 60 days and then sacrificed, and their liver, blood and gut feces were subsequently collected for testing. The liver tissue and fecal samples were subjected to RNA sequencing and full-length 16S rRNA sequencing analysis, respectively. Our results showed that the levels of serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) in the mice exposed to PVC-MPs were markedly higher than those in the control group, implying hepatic injury, as evidenced by hepatic histopathological changes. Moreover, the serum and hepatic triglyceride (TG) and total bile acid (TBA) levels were decreased after exposure to PVC-MPs. The RNA sequencing of mouse liver tissue identified a total of 1540 differentially expressed genes (DEGs) associated with 47 pathways, including the lipid metabolic pathway, oxidative stress, and the phosphoinositide 3 kinase (PI3K)/protein kinase B (Akt) signaling pathway, and these DEGs were enriched in the mouse livers. The full-length 16S rRNA sequencing analysis of the gut microbiota in mouse fecal samples revealed that PVC-MPs exposure decreased the relative abundance of probiotics and increased the abundance of conditionally pathogenic bacteria. In conclusion, chronic PVC-MPs exposure causes hepatotoxicity and gut microbiota dysbiosis in mice, and these findings provide new insight into the potential risks of PVC-MPs to human health.

* Corresponding author at: School of Forensic Medicine, Southern Medical University, South Shaitai Road #1023, 510515 Guangzhou, Guangdong, China.
E-mail address: qiaodf@163.com (D. Qiao).

¹ These authors have contributed equally to this work.

1. Introduction

Plastic pollution is currently widespread around the world. Statistics show that plastic is entering the environment at an alarming rate of 31.9 million tons a year, and it is estimated that there will be an additional 33 billion tons of plastic on the planet by 2050 (Rochman, 2018). In addition, plastics migrating into the environment cannot be easily degraded via physical strain, light and thermal oxidation processes and other environmental factors. Plastic can be continuously fragmented into microplastics (MPs, diameter less than 5 mm) or fragments with lower dimensions (less than 1 μm) known as nanoplastics. The ubiquitous nature of MPs in the environment results in inevitable human exposure to these particles (Prata et al., 2020). MPs have been detected in seafood, bottled water, beer, and household products (Banerjee and Shelver, 2021). Humans may be exposed to MPs through the digestive tract by ingesting food or drinking water or through the respiratory tract by inhaling air polluted with plastic dust (Prata et al., 2020). However, the consequences of the exposure of humans to MPs remain unclear. Polyvinyl chloride (PVC), as one of the most common MPs (Xu et al., 2020), is a raw material for chemical and consumer products, such as plastic pipes, disposable containers, automotive electronics, and toys (Jiang et al., 2020; Sadeghi et al., 2020). However, few studies have investigated the toxicity of PVC-MPs *in vivo*.

Previous experiments have shown that exposure to PVC products can lead to endocrine disruption and cancer (Hagmar et al., 1990). Mice exposed to 500 ppm PVC display maternal toxicity, as evidenced by decreases in body weight and litter size and increased mortality (John et al., 1981). The exposure of rats to 1000 mg/kg PVC can cause damage to the spermatogenic epithelium and spermatogenic disorder, resulting in a reduction in the testicular weight (Sadeghi et al., 2020). A previous study showed that the median lethal dose of PVC to rats is 5000 mg/kg (Archana et al., 2018). This concentration is high and is only observed in extreme environments. However, the long-term and low-dose effects of PVC-MPs on organisms are unclear. In our study, we selected a low PVC-MPs dose of 0.5 mg/day for 60 days to explore the effects of chronic PVC-MPs exposure on advanced animals. PVC exerts hepatotoxic and carcinogenic effects and induces renal, lung, and reproductive dysfunction (Sadeghi et al., 2020). The liver appears to be the primary target organ of PVC toxicity, which causes hepatocyte death and functional disorder (Jiang et al., 2020). Most previous studies have shown that in cases of exposure to PVC, the risk of hepatic angiosarcoma is 11 to 16 times higher than that in the normal population (Cave et al., 2010; Gennaro et al., 2008; Jiang et al., 2020; Wagoner, 1983). However, the mechanism of PVC-MPs-induced hepatotoxicity has not been fully elucidated.

The gut microbiota comprises a large number of microbial species living in the gastrointestinal tract and is involved in regulating many physiological functions of host organisms (Kataoka, 2016; Lynch and Pedersen, 2016). In fact, dysbiosis of the intestinal flora is usually associated with a variety of pathological changes, such as inflammatory bowel disease and abnormal behavior (Glenny et al., 2017; Weingarden and Vaughn, 2017). Increasing evidence demonstrates that the intestinal microbiota is a toxicological target of different types of environmental pollutants (Jin et al., 2017; Xia et al., 2018). Previous studies have indicated that MPs induce microbiota imbalance in the intestines of zebrafish and mice (Li et al., 2020; Qiao et al., 2019). However, the effect of PVC-MPs on the gut microbiota is unclear.

The purpose of this study was to explore the effects of chronic PVC-MPs exposure on mice. Histopathological examinations were conducted, and the serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) expression levels, serum and hepatic triglyceride (TG) levels, total bile acid (TBA) levels and gene transcript levels were measured to determine whether PVC-MPs cause hepatotoxicity. Furthermore, the composition of the intestinal microbiota was investigated. Our findings revealed that PVC-MPs increased the serum ALT and AST levels and decreased the TG and TBA levels in both serum and the liver, which are in accordance with the observed histopathological damage. This study provides new insights into the potential risks of PVC-MPs and the prevention and treatment of PVC-MP damage.

2. Materials and methods

2.1. Characterization of PVC-MPs

PVC-MPs (diameter of approximately 2 μm) were purchased from Huayu Technology Co., Ltd. (Guangzhou, China). The shape of the PVC-MPs was measured by scanning electron microscopy (Tescan, China) and atomic force microscopy (AFM) (SPM-9700, Shimadzu, Japan). The size and zeta potentials in distilled water were detected using a Zetasizer (Nano ZS90; Malvern, United Kingdom). The surface functional groups and structure of the PVC-MPs were measured by Fourier transform infrared (FTIR) spectroscopy (Nicolet iS10 instrument, Thermo Fisher, USA) and Raman microscopy (RM2000; Renishaw, Inc., United Kingdom).

2.2. Animals and treatment

Sixteen healthy male C57BL/6 mice (weighing 20–25 g, Laboratory Animal Center, Southern Medical University, Guangzhou, China) were individually housed in tub cages with food and water available *ad libitum*. All the animal procedures were approved by the National Institutes of Health Guide for the Care and Use of Laboratory Animals of Southern Medical University. The mice were habituated for 7 days before further experiments. The mice were randomly divided into two groups ($n = 8$) and received PVC-MPs (0.5 mg/day, oral gavage once a day) or saline (equal volume) for 60 days. The PVC-MPs were dispersed in saline and subjected to supersonic wave vibration for 20 min to ensure full suspension before use. The PVC-MPs concentrations were selected according to environmentally realistic concentrations of MPs in rivers (Eerkes-Medrano et al., 2015) and according to levels to which humans are exposed via consumer products (Dessi et al., 2021; Zhang et al., 2021).

2.3. Histopathology

Eight mice from each group were euthanized, and liver tissue collected from each group was immediately fixed with 4% paraformaldehyde solution for 24 h–48 h, dehydrated with different concentrations of alcohol, embedded in paraffin, sectioned to 3 μm and stained with hematoxylin and eosin (H&E) (Zhao et al., 2021). The histopathological changes in the liver were examined with a Leica DM5000B microscope (Leica, Germany).

2.4. Biochemical analysis

The serum ALT and AST levels were determined by enzymatic colorimetric assays with corresponding kits (Elabscience, USA) according to the manufacturer's instructions. The serum and hepatic TG, total cholesterol (TCH) and TBA levels were examined with corresponding kits (Nanjing Jiancheng, China) following the manufacturer's instructions. The levels of malondialdehyde (MDA, $\mu\text{mol}/\text{mg}$ protein) and superoxide dismutase (SOD, U/mg protein) in the liver were determined using special assay kits (Beyotime, China).

2.5. RNA sequencing (RNA-seq) and bioinformatics analysis

Total RNA from the liver was extracted with TRIzol (Takara, Japan). The quantity and quality of RNA were examined with a NanoDrop 2000 instrument (Thermo Fisher Scientific, USA). Approximately 10 μg of total RNA was used for transcriptome sequencing with an Illumina HiSeq 4000 (Ic-bio, Hangzhou, China) according to the recommended protocol. The differentially expressed genes (DEGs) showing statistical significance were obtained from the sequencing results based on the following criteria: $|\log_2(\text{fold change})| < 1$ and p value < 0.05 . The DEGs were introduced into the Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) databases for annotation analysis using online tools, such as <https://www.omicstudio.cn/tool> and <http://www.bioinformatics.com.cn>.

2.6. Quantitative real-time PCR (qRT-PCR) analysis

To test the accuracy of the RNA sequencing results and validate the expression of the genes, nine DEGs (*Moxd1*, *Rnf138rt1*, *Mt2*, *Lcn2*, *Cyp2b9*, *Pcdhga5*, *Serinc4*, *Gpat3* and *Ppar-γ*) were randomly selected and the mRNA expression levels of these genes were determined by qRT-PCR (Chen et al., 2019b). The mRNA expression levels of *Nrf2*, *Ho-1*, *SOD* and *NQO1* were also determined by qRT-PCR. The GAPDH gene was used as an internal reference gene. Subsequently, cDNA was synthesized using the HiScript II 1st Strand cDNA Synthesis Kit (Vazyme, Nanjing, China). The sequences of the primers used for the nine DEGs and *Nrf2*, *Ho-1*, *SOD* and *NQO1* were designed by and purchased from Sangon Biotech Incorporation (Shanghai, China), as shown in Table S1, and qRT-PCR was conducted with a SYBR Green system (Vazyme, Nanjing, China).

2.7. Western blotting analysis

Protein expression in liver tissue was analyzed by Western blotting (WB) as previously described (Chen et al., 2020). Proteins were separated by 10% sodium dodecyl sulfide (SDS)-polyacrylamide gel electrophoresis and electrophoretically transferred to polyvinylidene fluoride (PVDF) membranes. The transferred blots were incubated with rabbit anti-β-actin (ABclonal, USA), rabbit anti-P53 (Abcam, UK), rabbit anti-PTEN (CST, USA), rabbit anti-PI3K (Proteintech, USA), rabbit anti-p-Akt (CST, USA), rabbit anti-Akt (CST, USA), rabbit anti-Bax (CST, USA), rabbit anti-cleaved caspase-3 (CST, USA), *Nrf2* (CST, USA) and *Ho-1* (CST, USA) overnight at 4 °C. Horseradish peroxidase (HRP)-coupled goat anti-rabbit/mouse IgG (Bioss, China) was utilized as the secondary antibody.

2.8. DNA extraction, qPCR amplification and full-length 16S rRNA gene sequencing

All mouse fecal sample collection and DNA extraction procedures were performed using the Magnetic Soil and Stool DNA Kit (TIANGEN, China) according to the user manual. Full-length 16S rRNA analysis was performed using the primers 27 F (5'-AGAGTTTGATCMTGGCTCAG-3') and 1492 R (5'-ACCTTGTTACGACTT-3'). The microbiota analyses were performed on the BMKCloud platform (www.biocloud.net). The effect tags were clustered at the 97% similarity level, and operational taxonomic units (OTUs) were obtained by USEARCH (Version 10.0) in QIIME software (Version 1.8.0). The classification annotation of OTUs was performed according to the Silva ribosomal RNA database (<http://www.arb-silva.de>). Related indexes such as alpha and beta diversity indexes were analyzed using QIIME software in BMK Cloud (www.biocloud.net).

2.9. Statistical analysis

All experiments involving biochemical measurements and WB were performed in triplicate. SPSS 20.0 (SPSS, Chicago, IL, USA) was used for the statistical analyses. All values are reported as the means ± standard errors of the mean (SEMs). Student's *t*-test was utilized to test the significance of the differences between two groups. The bacterial relative abundances were analyzed using the Kruskal-Wallis test with false discovery rate (FDR) correction for multiple testing. The correlation analysis between the relative abundance of bacteria and liver injury was investigated by Spearman correlation analysis. All statistical analyses were performed using GraphPad Prism 9. A value of *p* < 0.05 was considered to indicate statistical significance.

3. Results

3.1. Characteristics of PVC-MPs

The properties of the PVC-MPs are shown in Fig. 1 and Table 1. The morphology of the PVC-MPs used in our study was tested by scanning electron microscopy (Fig. 1A) and AFM (Fig. 1B and Fig. S1A), and the results

indicated that the PVC-MPs were characterized as particles with an approximately circular morphology and a rough surface. The two-dimensional (Fig. 1B) and three-dimensional (Fig. S1A) AFM images revealed that the surface of the PVC-MPs was rough and regularly distributed and the Ra value of the surface roughness is 1.158 nm. The FTIR spectral analysis of the PVC-MPs demonstrated that peaks appeared at 1435.47 cm⁻¹ and 1387.82 cm⁻¹ (stretching vibration of methylene) and at 2950.55 cm⁻¹ and 2993.58 cm⁻¹ (stretching vibrations of C—H bonds in the methyl group) (Fig. S1B). The FTIR analysis confirmed that some chemical functional groups did not belong to PVC, which suggested that the PVC contained a small number of ester additives. The FTIR analysis provided information on the functional groups of the PVC-MPs, and the Raman spectral results (Fig. S1C) were basically consistent with the FTIR analysis. The average particle size (Fig. 1C) and zeta potential (Fig. S1D) of the PVC-MPs were 2.08 ± 0.21 μm and -37.6 ± 7.57 mV, respectively. Taken together, these results indicated that the PVC-MPs exhibited favorable stability and monodispersity and could be applied in mammals.

3.2. Effects of exposure to PVC-MPs on hepatotoxicity

The experimental design used to examine exposure to PVC-MPs and saline is shown in Fig. 2A. The histopathological analysis revealed prominent histological changes in the livers of mice after exposure to PVC-MPs (Fig. 2B). The PVC-MPs significantly damaged the microstructure of hepatic cells by causing cytoplasmic damage in the test group compared with the control group, and the effects manifested as hepatocyte ballooning, cell vacuolization and dilated intercellular space. The serum expression of ALT and AST was positively correlated with the degree of liver injury (Fig. 2C). The ALT and AST contents were increased after PVC-MP treatment. Moreover, the levels of serum TG, TCH and TBA were decreased after the administration of PVC-MPs (Fig. 2D). Among the studied hepatic indexes, the levels of hepatic TG and TBA, but not the hepatic TCH levels, were prominently decreased after PVC-MPs exposure (Fig. 2E).

3.3. Principal component analysis (PCA) and screening of DEG correlations

The RNA-seq results were simplified by PCA. This analysis showed that the samples from the PVC-MPs group (red circle) and those from the control group (blue circle) were clustered (Fig. 3A) and revealed a higher difference between the groups and lower variation within groups in the current study. The volcano plot showed that 563 and 977 DEGs showed markedly upregulated and downregulated expression after the PVC-MPs treatment compared with the levels found in the control group (Fig. 3B).

The expression profiles of the top 40 DEGs identified from the RNA-seq analysis are shown in the heatmap in Fig. 3C. To test the accuracy of the RNA-seq results, nine genes (*Moxd1*, *Rnf138rt1*, *Mt2*, *Lcn2*, *Cyp2b9*, *Pcdhga5*, *Serinc4*, *Gpat3* and *Ppar-γ*) were randomly selected for qRT-PCR analysis. The results showed that the PVC-MPs administration upregulated the expression of the DEGs *rnf138rt1*, *Moxd1*, *Mt2* and *Lcn2* and downregulated the expression of the DEGs *Cyp2b9*, *Pcdhga5*, *Serinc4*, *Gpat3* and *Ppar-γ*, which was in line with the RNA-seq results (Fig. 3D-E).

3.4. Functional annotation and classification of DEGs

To test the biological functions of the DEGs, GO and KEGG pathway analyses were performed with all the DEGs. The GO annotations of these genes can be divided into three independent categories according to their functions and pathways: biological process (BP), cellular component (CC), and molecular function (MF). As shown in Fig. 4A, membrane and protein binding were the most representative functions. The KEGG pathway analysis also provided insights into the relevant pathways (Fig. 4B). The DEGs identified from the two groups were further annotated into five classifications: organismal systems, metabolism, human diseases, environmental information, and cellular processes.

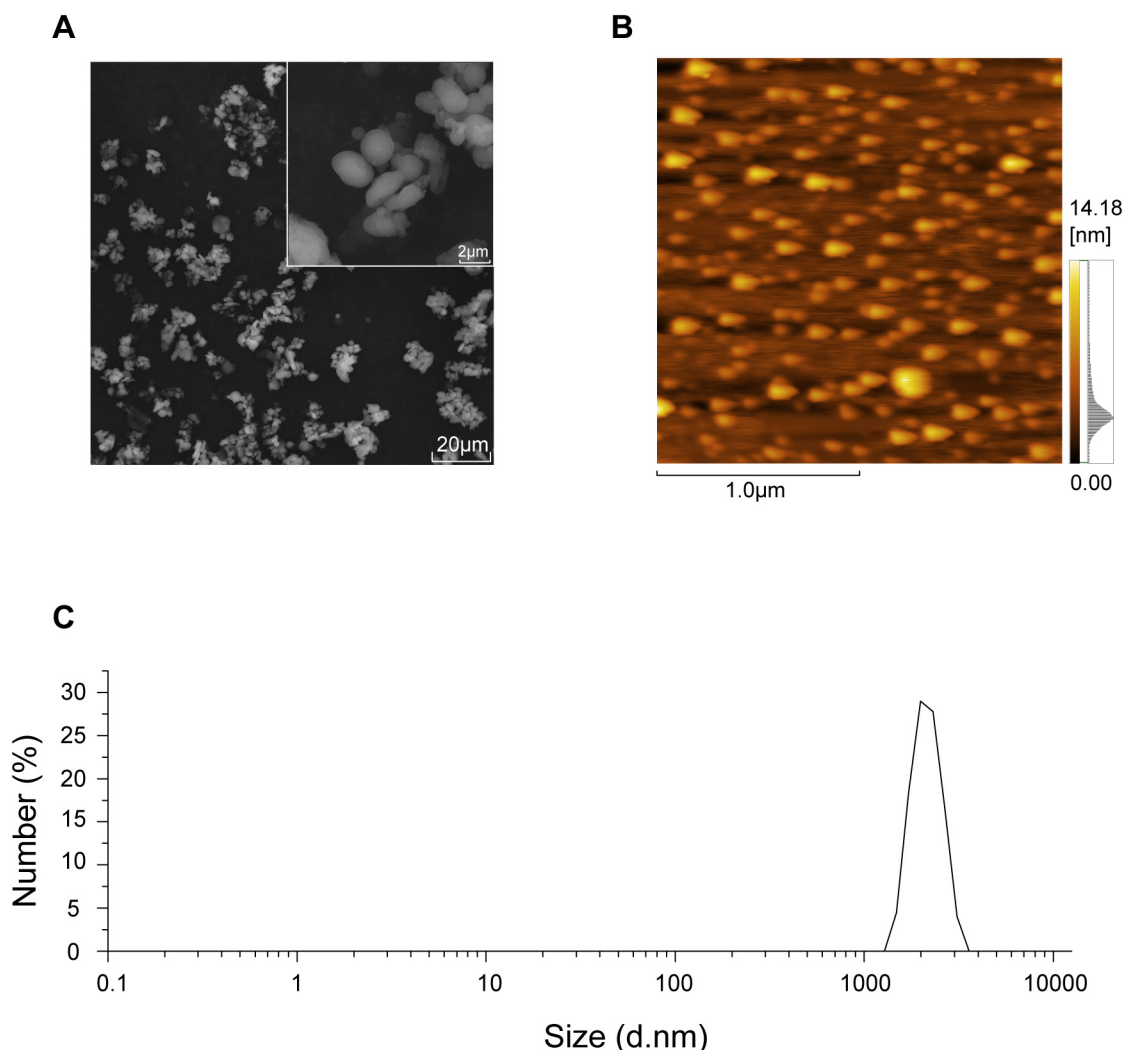


Fig. 1. Characterization of the 2.0- μm polyvinyl chloride (PVC) microplastics (MPs) used in this study. (A) Image of PVC MPs under a scanning electron microscope ($\times 2000$ and $\times 20000$ magnification, respectively). (B) Two-dimensional AFM image. (C) Size distribution by intensity. Each examination was performed with at least three independent replicates.

3.5. PVC-MPs induced apoptosis via the PI3K/Akt pathway

The KEGG analysis showed that the PI3K/Akt pathway was markedly altered in the PVC-MPs treatment group (Fig. 4B). Subsequently, this signaling pathway was explored by WB, and the results showed that exposure to PVC-MPs notably increased the protein expression levels of phosphatase and tensin homolog (PTEN), P53, PI3K, phosphorylated Akt (p-Akt), Akt and its downstream apoptosis-related proteins, cleaved caspase3 and Bax compared with the levels found in the control group (Fig. 5).

3.6. Expression of oxidative stress-related proteins

Exposure to PVC-MPs induced an increase in the MDA levels and markedly decreased the antioxidant enzymatic activity of SOD compared with

those in the control group (Fig. 6A). Nrf2, as a crucial transcription factor of the antioxidant response, regulates the gene expression of antioxidant response elements to cooperatively activate a series of detoxification/defense genes and participates in regulating the expression of a series of detoxification and antioxidant defense genes in the liver (Taguchi and Kensler, 2020). Herein, we tested the classical antioxidant pathway, the Nrf2/Ho-1 signaling pathway. The mRNA expression levels of Nrf2, Ho-1, NQO1 and SOD and the protein expression levels of Nrf2 and Ho-1 after PVC-MPs exposure were significantly lower than those in the control group (Fig. 6B-C).

3.7. Effects of PVC-MPs exposure on changes in the gut microbiota

Full-length 16S rRNA sequencing generated 8275–11,630 effective tags for each sample. The alpha diversity was estimated according to richness and diversity indexes. The results revealed no significant differences in richness (Chao, $p = 0.884$; Ace, $p = 0.773$) or diversity (Shannon, $p = 0.391$; Simpson, $p = 0.391$) among the groups (Fig. S2). With respect to beta diversity, the differences in the bacterial composition among groups were visualized by hierarchical clustering analysis (HCA) and principal coordinate analysis (PCoA). HCA could separate the groups based on the Bray–Curtis method at the OTU level (Fig. 7A). The PCoA results showed that the microbial composition of the PVC-MPs exposure group was notably different from that of the control group (Fig. 7B). Exposure to PVC-MPs changed

Table 1

Physicochemical properties of PVC MPs.

Parameter	PVC MPs
Form	powder
Particle shape	approximate spherical morphology
Particle surface	rough ($R_a = 1.158 \text{ nm}$)
Particle size	$2.08 \pm 0.21 \mu\text{m}$
Zeta potential	$-37.6 \pm 7.57 \text{ mV}$

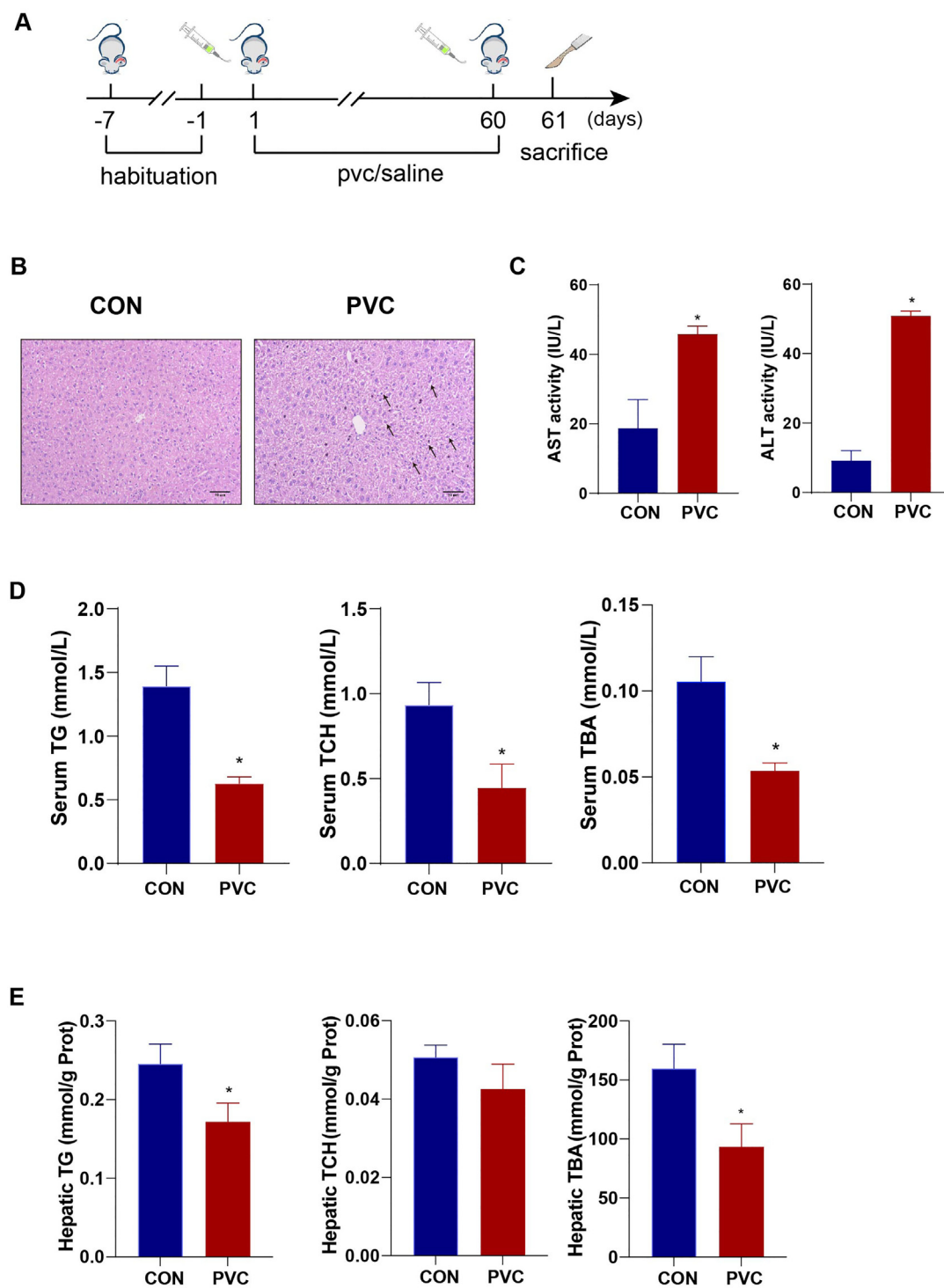


Fig. 2. Histopathological and biochemical assessments of liver injury. (A) Experimental design used to examine PVC-MPs exposure. (B) PVC-MPs induced hepatocyte ballooning compared with the control group. Scale bar (black), 10 μ m. The black arrows indicate hepatocyte ballooning. (C) PVC-MPs significantly enhanced the levels of ALT and AST in plasma relative to the control group. Serum (D) and hepatic (E) triglyceride (TG), total cholesterol (TCH) and total bile acid (TBA) expression levels. The data for each group are presented as the means $\pm$ SEMs ($n = 8$). * $p < 0.05$, compared with the control group.

the percentages of the gut microbiota at the phylum (Fig. 7C), genus (Fig. 7D), and species (Fig. 7E) levels.

In particular, lower *Bacteroidetes* ($p = 0.0055$) and higher *Verrucomicrobia* ($p = 0.0107$), *Firmicutes* ($p = 0.0194$), *Proteobacteria* ($p = 0.0217$), and *Patescibacteria* ($p = 0.0405$) levels were observed in the PVC-MPs-treated group compared with the control group (Table S2). In addition, exposure to PVC-MPs augmented the relative abundances of *Parasutterella*, *Roseburia*, *Dubosiella* and *Akkermansia* but reduced those of

Aerosphaera, *Aquamicrobium*, *Brevibacterium*, and *Ruminiclostridium_5* at the genus level (Table S3). Furthermore, the species-based classification of gut microbiota communities revealed notably higher abundances of *uncultured_bacterium_g_Parasutterella*, *Firmicutes_bacterium_M10-2*, *Brevibacterium_epidermidis*, *Staphylococcus_lentus*, and *Psychrobacter_sanguinis* and lower abundances of *uncultured_bacterium_f_Muribaculaceae* and *Paenalcigenes_hominis* in the PVC-MPs group compared with the control group (Table S4). The results of the cladogram and linear discriminant

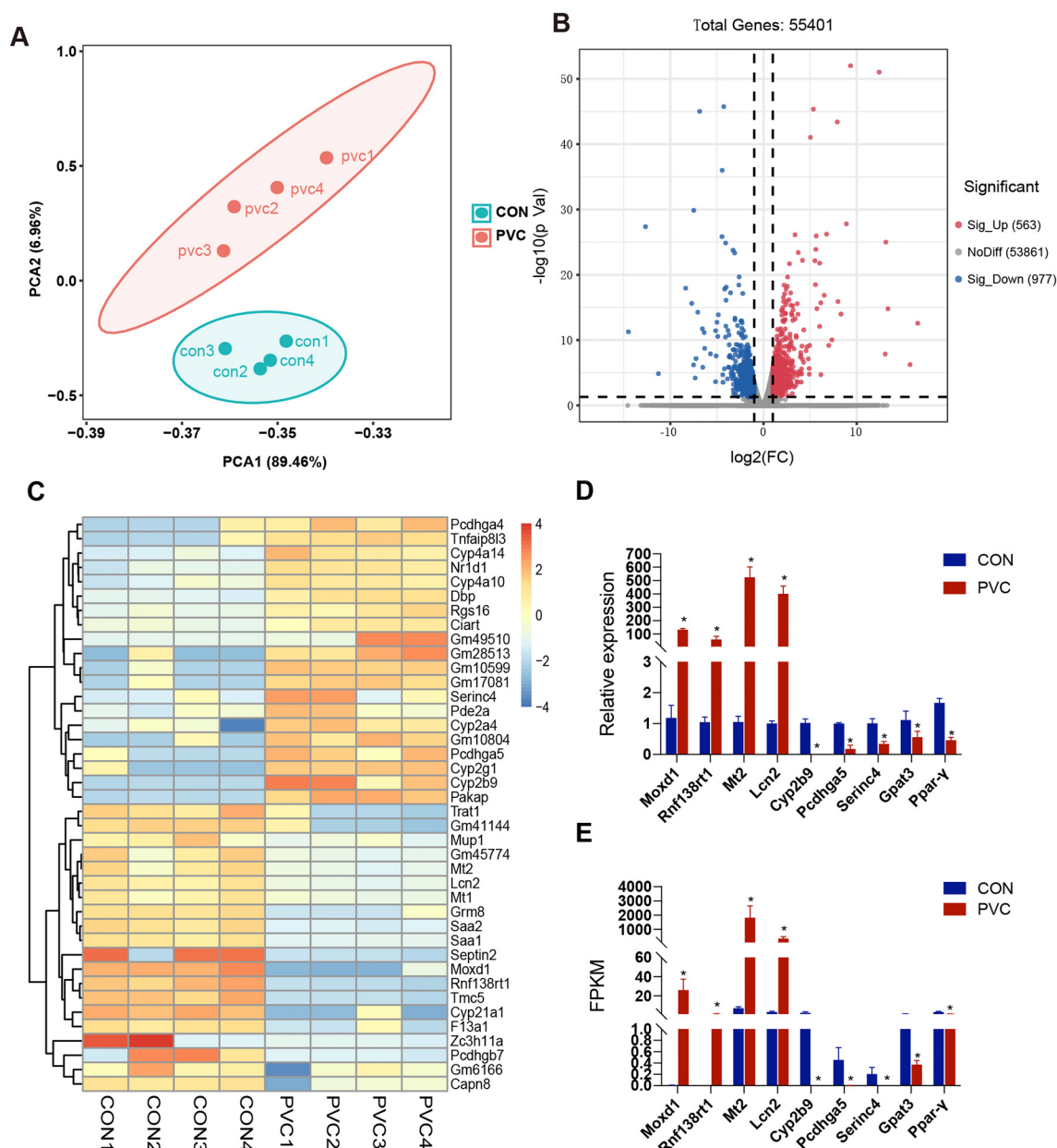


Fig. 3. Analysis of differentially expressed genes (DEGs) between the test groups. (A) According to the principal components, the intragroup samples were clustered, and the two groups were distinctly separated. (B) DEGs between the control and the PVC-MPs-treated group are represented on a volcano plot map ($\log_{10} p$ value vs. $\log_2 FC$) ($n = 4$). The red and blue dots indicate transcripts with positive and negative change values, respectively, which indicated the upregulation and downregulation of DEGs in the PVC-MPs-exposed group. (C) Gene heatmap comparing the PVC-MPs-treated and -untreated samples. The red or positive numbers indicate upregulation, and the blue or negative numbers indicate downregulation. (D-E) The qRT-PCR results were consistent with the RNA-seq results. The data for each group are presented as the means $\pm$ SEMs ($n = 4$). * $p < 0.05$, compared with the control group.

analysis (LDA) showed increases in *Firmicutes*, *Staphylococcus*, *Erysipelotrichales*, *Akkermansia*, and *Dubosiella* in the PVC-MPs group relative to the control group (Fig. 7F–G).

3.8. Analysis of the correlations between the gut microbiota and liver injury indexes

The correlations between the gut microbiota and lipid metabolism-related DEGs identified by RNA-seq analysis were explored by Spearman's correlation analysis (Fig. 8A). The results from the genus-level analysis showed that the change in the abundance of *Akkermansia* was positively correlated with the levels of cholesterol metabolism (Apoa4 and Acat2) and bile secretion (Abcb1b, $p < 0.05$). The change in the abundance of *Staphylococcus* was negatively associated with the

DEGs involved in lipid metabolism processes such as fatty acid degradation (Cyp4a14 and Cyp4a10, $p < 0.05$), cholesterol metabolism (Lpl, $p < 0.05$) and fatty acid elongation (Acot1, Acot3 and Acot4, $p < 0.05$). In addition, Spearman's correlation analysis was also performed to discover the correlations among the gut microbiota and biochemical indicators, such as serum ALT and AST, hepatic and serum TG, and TBA and TCH levels. As shown in Fig. 8B, the results revealed a stronger correlation between the gut microbiota and biochemical indexes. For example, serum AST exhibited positive correlations with serum ALT ($r = 0.85$, $p < 0.05$), *Akkermansia* ($r = 0.96$, $p < 0.05$), and *Staphylococcus* ($r = 0.2$, $p < 0.05$) but negative correlations with serum TG ($r = -0.64$, $p < 0.05$), serum TCH ($r = -0.85$, $p < 0.05$), serum TBA ($r = -0.85$, $p < 0.05$), hepatic TBA ($r = -0.71$, $p < 0.05$), and *Muribaculum* ($r = -0.70$, $p < 0.05$).



Fig. 4. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) analyses of DEGs between the PVC-MPs-treated and control groups. (A) The functions of the identified genes were categorized into three main categories in the GO analysis: biological process, cellular component, and molecular function. The left y-axes indicate the number of genes in a category, respectively. (B) Top 30 enriched KEGG pathways ($n = 4$). The x-axis indicates the percentage of enriched DEGs, and the y-axis indicates the second-level GO term.

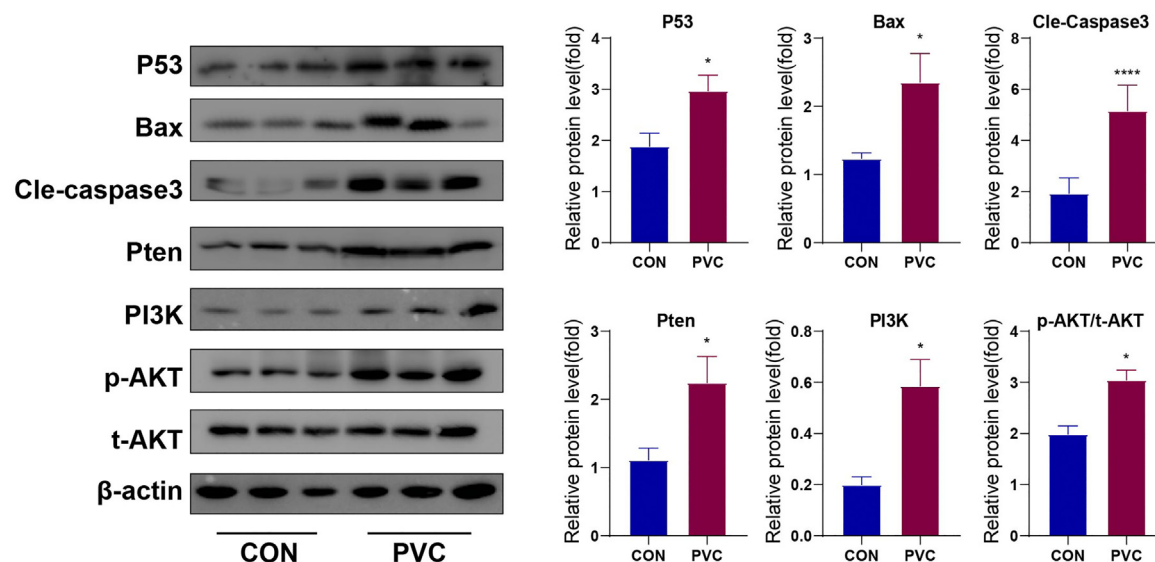


Fig. 5. Western blotting analysis of the protein expression of PTEN, p53, PI3K, p-Akt, Akt, cleaved caspase 3, and Bax. Treatment with PVC-MPs significantly increased the expression of PI3K, p-Akt, PTEN, p53, cleaved caspase 3 and Bax compared with the levels in the control group. The data for each group are presented as the means $\pm$ SEMs ($n = 3$). * $p < 0.05$, compared with the control group.

4. Discussion

Plastic is prevalently used in our daily life, and its health impacts on humans and other organisms have become a research hotspot. PVC is a common MP environmental pollutant that has recently been proven to be toxic to the intestinal tract and liver of aquatic organisms and terrestrial mammals (Espinosa et al., 2019; Jiang et al., 2020). In the present study, we explored the associations between exposure to PVC-MPs (0.5 mg/day) for 60 days and the levels of hepatotoxicity and the gut microbiota. In particular, we found that chronic PVC-MPs exposure significantly induced hepatic injury, which induced cell death according to hepatic histopathology, increases in the serum ALT and AST levels and decreases in the serum and hepatic TG and TBA levels, apoptosis-related PI3K/Akt and oxidative stress activation related to the Nrf2/Ho-1 signaling pathway. Moreover, we observed that long-term PVC-MPs exposure caused gut microbiota dysbiosis in mice. The concentrations of PVC-MPs used in this study were based on the PVC-MPs content in aquatic ecosystems (Eerkes-Medrano et al., 2015; Santana et al., 2018) and in the levels to which humans are exposed via consumer products (Dessi et al., 2021; Zhang et al., 2021). Therefore, a dietary concentration of 0.5 mg/day could be considered a valid reflection of a contaminated area.

Histopathology is a good approach for detecting tissue damage. In this study, we observed vacuolation of hepatocytes and dilated intercellular space in the PVC-MPs-treated group for 60 days, which indicated that chronic exposure to PVC-MPs induced liver injury and affected the normal function of hepatocytes, and these findings were consistent with the upregulated expression of AST and ALT in serum, which are the main parameters used as important indexes of liver diseases (Iheanacho and Odo, 2020; Jung et al., 2003). Espinosa et al. reported similar liver histopathological findings, including an increase in hepatocyte vacuolation and loss of parenchyma organization in PVC-MPs-fed *Dicentrarchus labrax* (Espinosa et al., 2019), which was consistent with our results. Furthermore, we found that the TG and TBA levels in both serum and liver tissue were decreased after PVC-MPs exposure, which indicated that long-term exposure to PVC-MPs also led to disordered TG and TBA metabolism. TG plays an important role in providing and storing energy and fixing and protecting internal organs (Lu et al., 2018b). The decreased expression of TG in the liver leads to liver dysfunction and damage to hepatocytes and may lead to liver lipid metabolism disorder.

To date, most studies have focused on characterizing morphological abnormalities, but few studies have explored the potential molecular

mechanisms affecting these changes (Besseling et al., 2018; Shen et al., 2019). For this reason, transcriptomic and bioinformatic analyses were performed in this study to predict the toxicity and evaluate the mechanisms of action associated with PVC-MPs exposure in mice. A total of 1540 DEGs and 47 pathways were identified in mice after PVC-MPs treatment. KEGG and GO analyses showed that lipid metabolic and oxidative stress-related signaling pathways were enriched after exposure to PVC-MPs. Among all of these signaling pathways, most genes were enriched in cancer pathways after PVC-MPs exposure, which indicated that chronic PVC-MPs exposure poses a potential cancer risk. These DEGs were further analyzed using a series of bioinformatics methods, which appeared to be helpful for understanding the molecular mechanism influencing the expression of DEGs after PVC-MPs administration.

In this study, a KEGG pathway analysis of the RNA-seq results identified pathways related to PVC-MPs-induced hepatotoxicity and showed that the PI3K/Akt signaling pathway was enriched after PVC-MPs exposure. The activation of Akt phosphorylation provokes downstream mTOR-induced inhibition of apoptosis and enhanced cell proliferation (Hay, 2011). Herein, PVC-MPs exposure upregulated PI3K and p-Akt expression, which indicated that PVC-MPs treatment induced activation of the PI3K/Akt signaling pathway. In addition, Akt activation is necessary for P53 stability and accumulation (Abraham and O'Neill, 2014). Our results suggested that the expression levels of the apoptosis-related proteins P53, cleaved caspase 3 and Bax were also markedly increased in the mouse liver after PVC-MPs exposure, which was consistent with the observed hepatic histopathological changes. P53 eliminates the PI3K/Akt signaling pathway and thereby negatively regulates the mTOR-mediated inhibition of apoptosis (Grinkevich et al., 2009). Therefore, PI3K/Akt/mTOR-inhibited apoptosis may be recovered, which is consistent with the finding that P53 promotes apoptosis by activating the caspase signaling pathway. In addition, the accumulation of P53 enhances the transcription of PTEN, which is the main antagonist of the PI3K/Akt pathway (Pan et al., 2018). P53 combines with the PTEN promoter to form a positive feedback loop and protect P53 from degradation (Chen et al., 2005). In the current study, exposure to PVC-MPs markedly increased the expression level of PTEN. Therefore, the PI3K/Akt pathway activated by PVC-MPs may accelerate the accumulation of P53 and the increase in PTEN mediated by P53, thus offsetting this pathway. In addition, P53 has previously been shown to cause apoptosis, which may play a vital role in PVC-MPs-induced hepatotoxicity. The results from the observations of hepatic histopathology and molecular protein levels indicated that PVC-MPs induced hepatic injury.

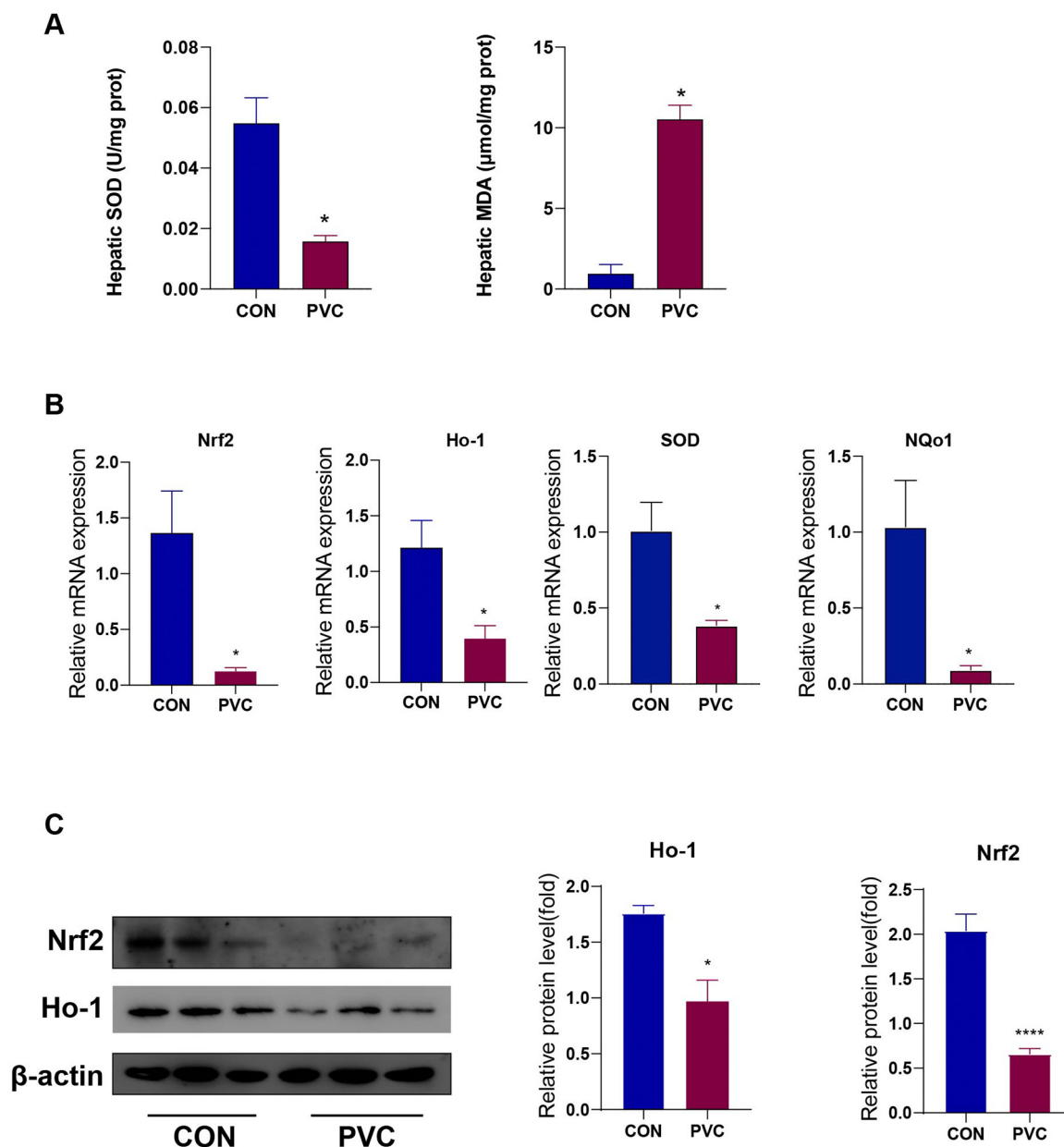


Fig. 6. Effects of PVC-MPs on oxidative stress in mouse livers. (A) Exposure to PVC-MPs decreased the protein expression level of SOD and increased the MDA content compared with the control group. (B) Nrf2, Ho-1, SOD and NQO1 mRNA expression and (C) Nrf2 and Ho-1 protein expression were decreased after PVC-MPs exposure. The data for each group are presented as the means $\pm$ SEMs ($n = 3$). * $p < 0.05$, compared with the control group.

In the current study, several significant pathways were found to be altered (drug metabolism-cytochrome P450, arachidonic acid metabolism and glutathione metabolism), and these pathways are involved in the defense against oxidative stress (Balboa and Balsinde, 2006; Chen et al., 2019a). These results are in line with our data and previous studies, which indicate that PVC-MPs toxicity can lead to oxidative stress (Archana et al., 2018; Iheanacho and Odo, 2020). Therefore, oxidative stress defense appears to be an important protective response of mammals to microplastic toxicity. Under certain pathological conditions, excessive activation of oxidative stress can induce cell death and tissue damage (Li et al., 2015). MDA is a lipid oxidation product, and its content can reflect the degree of lipid peroxidation in the body (Yu et al., 2018). Our results showed that PVC-MPs led to an increase in MDA and a decrease in SOD in liver tissue. Nrf2, as one of the vital transcription factors in the antioxidant defense system (Zhang et al., 2019), can increase the expression of Ho-1 to protect cells/tissues from oxidative stress (Yao et al., 2019). In this study, we found that Nrf2, Ho-1 and NQO1 expression were

significantly decreased after PVC-MPs treatment, which indicated that the regulation of Nrf2/Ho-1 might be one of the mechanisms involved in PVC-MPs-induced oxidative stress. These results suggested that PVC-MPs induced oxidative stress in the liver. In the future, further in-depth research on oxidative stress is needed to understand how oxidative stress affects body health.

The gut microbiota plays a vital role in biological barriers, host metabolism, physiology, and immune system development (Gill et al., 2006). The intestinal microbiota is dominated by bacteria, and more than 90% of the included species belong to *Bacteroides* and *Firmicutes*. In this study, PVC-MPs administration decreased the abundance of *Bacteroides* and increased that of *Firmicutes* in the mouse intestinal microbiota. Specifically, PVC-MPs decreased the relative abundances of several probiotics, including *Muribaculaceae*, *Prevotellaceae* and *Enterorhabdus*, which are associated with short-chain fatty acid (SCFA) production (Liang et al., 2021; Wang et al., 2021). SCFAs play vital roles in regulating energy metabolism and maintaining the balance of the intestinal mucosal barrier in the intestine

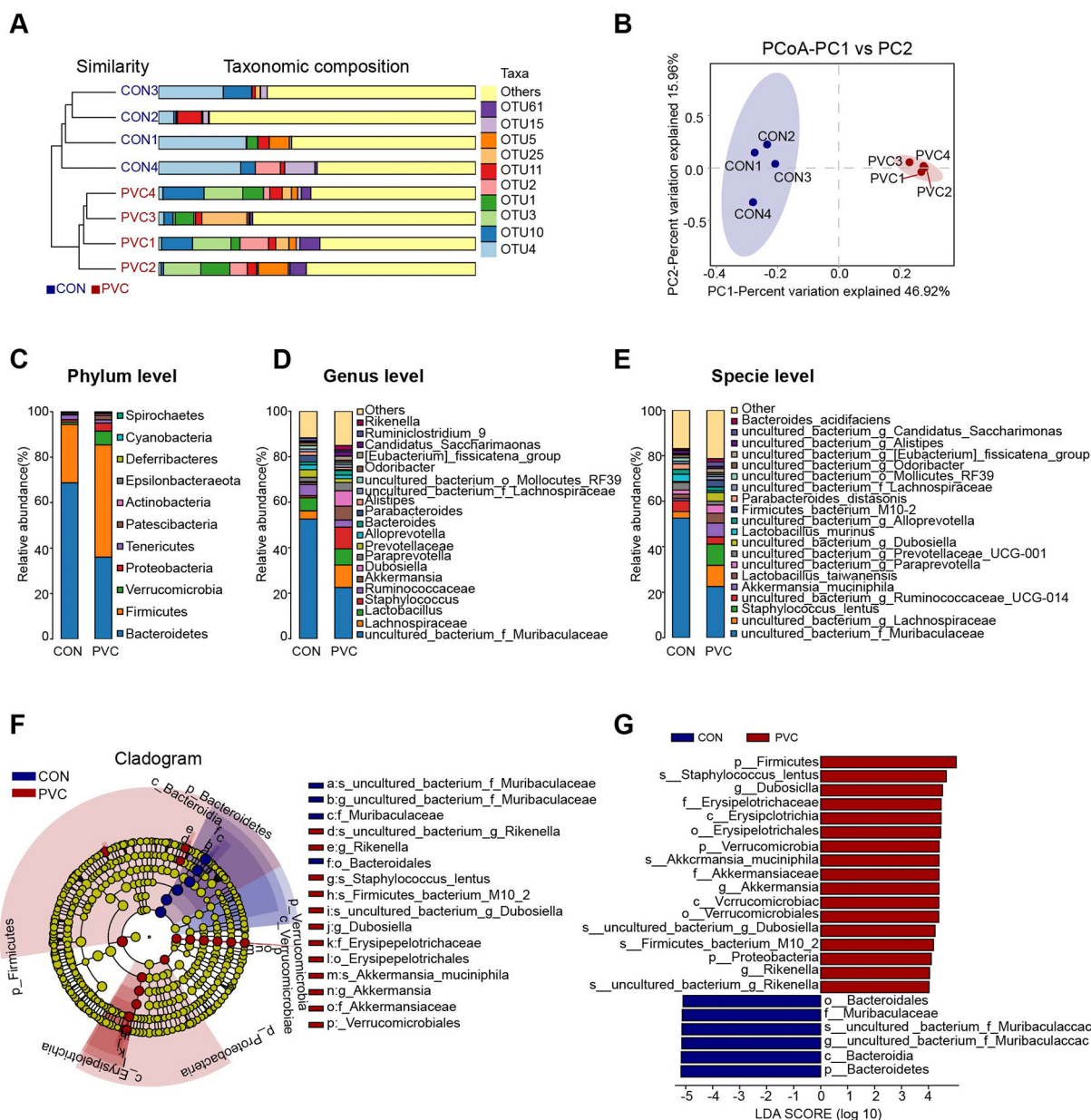
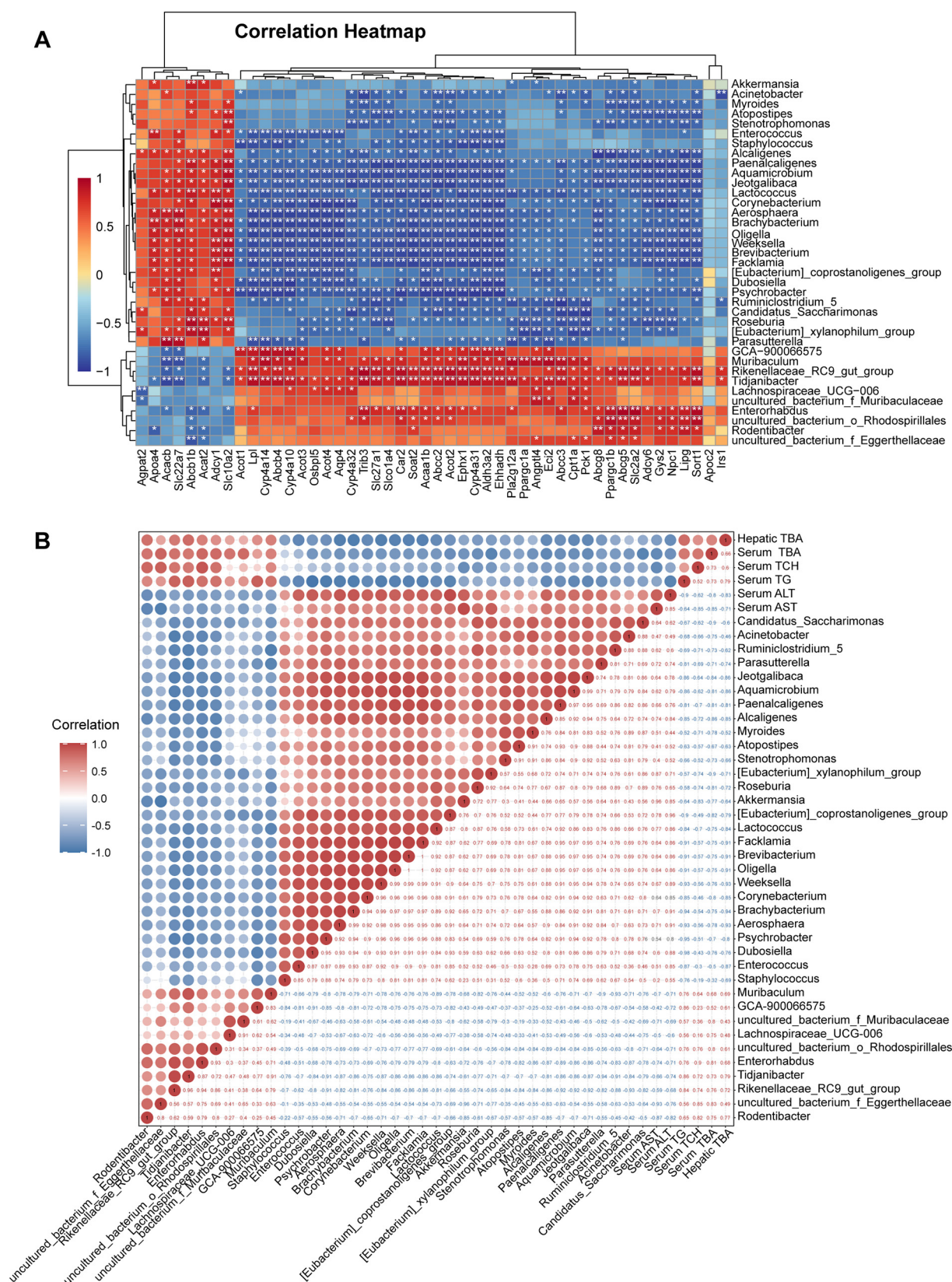


Fig. 7. Microbiome alterations induced by PVC-MPs exposure. (A) Hierarchical clustering tree based on the Bray–Curtis method and (B) principal coordinate analysis (PCoA) based on the Bray–Jaccard method at the OTU level. Bacterial community compositions at the phylum (C), genus (D), and species (E) levels. The data for each group are described as the means $\pm$ SEMs. The microbial abundance was calculated as a percentage of the total bacterial taxon within each sample. LeifSe cladogram (F) and LDA (G) value distribution histogram showing the different abundant bacterial taxa between the control and PVC-MPs groups. The red and blue colors indicate distinct bacterial taxa in the control and PVC-MPs groups, respectively, whereas the yellow color denotes no significant differences between the groups. LeifSe analysis of the phylum to species classification levels using an all-against-all (stricter) comparison strategy. Taxa with LDA values greater than three are presented ($n = 3$).

(Morrison and Preston, 2016). Exposure to PVC-MPs reduced the abundance of SCFA-producing bacteria, which suggested that PVC-MPs disrupted the balance of the intestinal mucosal barrier. The reduction of probiotics leads to the reduction of gut-protective factors, which makes the gut mucosal barrier vulnerable to damage. Furthermore, PVC-MPs exposure increased the abundance of conditionally pathogenic bacteria, such as *Parasutterella*, *Blautia*, *Staphylococcus* and *Mucispirillum* (Chen et al., 2018; Herp et al., 2021; Kong et al., 2019). Previous studies have shown that *Staphylococcus* overgrowth is associated with bowel disease due to superantigen-induced inflammation and damage (Collado et al., 2008). The results from the correlation analysis showed that the abundance of *Staphylococcus* was significantly increased and that of *Parabacteroides* was decreased in the PVC-MPs group. The changes in *Staphylococcus*

abundance were positively correlated with serum AST and ALT and negatively correlated with the levels of serum TG, TBA and TCH and hepatic TBA, which indicated that changes in the *Muribaculaceae* abundance were related to liver injury and lipid metabolism (Fig. 8B). Taken together, our results indicated that PVC-MPs induced gut microbiota dysbiosis, as manifested by an increase in conditional pathogenic bacteria and a decrease in beneficial probiotics, and destroyed the balance of the intestinal barrier.

The liver is an extremely important organ and metabolic center in vertebrates (Nguyen et al., 2008). Previous studies have shown that the intestinal microbiota is involved in host energy metabolism (Chopyk and Grakoui, 2020). The serum and hepatic TG and TBA levels were significantly decreased in the PVC-MPs exposure group. Furthermore, hepatic lipid metabolism is mainly regulated by certain genes in the PPAR family



(Nguyen et al., 2008), which were enriched after PVC-MPs administration. PPAR γ , as an important transcription factor, plays an important role in liver lipid metabolism (Coleman and Lee, 2004; Lu et al., 2018a). The RNA-seq and qRT-PCR results showed that PVC-MPs exposure induced a decrease in hepatic PPAR γ at the transcriptional level, which was consistent with the expression of TG and TBA. Emerging evidence suggests the existence of a strong relationship between the intestinal microbiota and the liver; specifically, some products of metabolites from the intestinal microbiota indirectly affect the function of the liver (Compare et al., 2012). Our results showed that *Akkermansia* and *Muribaculum* were positively correlated with the expression of DEGs associated with cholesterol metabolism and bile secretion, whereas *Staphylococcus* was negatively associated with the expression of DEGs involved in lipid metabolism, such as fatty acid degradation and cholesterol metabolism. Furthermore, *Erysipelotrichaceae* and *Akkermansia*, which are related to SCFA and bile acid metabolism, were changed after PVC MP exposure (Fuhren et al., 2021; Shaidulloev et al., 2021). Some recent studies have also shown that polystyrene MPs can accumulate in the liver and affect liver metabolism (Deng et al., 2017; Lu et al., 2018a), which is partially consistent with our results. The results from the correlation analysis showed that the observed change in *Muribaculaceae* was negatively correlated with the serum ALT and AST levels but positively correlated with the serum TG, TBA, and TCH levels, and hepatic TBA levels, whereas the opposite correlations were obtained for *Staphylococcus* (Fig. 8B). The changes in these commensal bacteria led to a decreased content of bile acids, which is related to disordered lipid metabolism. Based on these results, we speculate that the disorder of liver lipid metabolism caused by PVC-MPs may be related to gut microbiota dysbiosis.

5. Conclusion

In summary, the effects of PVC-MPs on hepatic injury and gut microbiota dysbiosis in mice were analyzed. We found that PVC-MPs induced hepatotoxicity, including hepatic lipid metabolism disorder, abnormal gene regulation at the transcriptional level, and the upregulation of ALT and AST, and these effects were consistent with the observed histopathological abnormalities. Moreover, exposure to PVC-MPs changed the structure of the gut microbiota and affected the balance of the intestinal mucosal barrier. Therefore, the results from the present study provide new insights into the treatment and prevention of PVC-MPs-induced hepatotoxicity and gut microbiota dysbiosis *in vivo*.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2022.155984>.

CRediT authorship contribution statement

Xuebing Chen: Methodology, Investigation, Writing – original draft, Writing – review & editing. **Jingshen Zhuang:** Investigation, Validation, Writing – original draft. **Qianling Chen:** Investigation, Validation. **Luyao Xu:** Investigation, Validation. **Xia Yue:** Conceptualization, Supervision, Resources, Writing – review & editing. **Dongfang Qiao:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank Biomarker Technologies Co., Ltd. (Beijing, China), for the assistance provided with the full-length 16S rRNA sequencing and Lianchuan Technologies Co., Ltd. (Hangzhou, China), for the assistance provided with RNA sequencing. We thank American Journal Experts for their linguistic assistance during the preparation of this manuscript.

Data availability

All data needed to evaluate the conclusions described in the paper are presented in the manuscript and/or the Supplementary Materials. Additional data are available from the corresponding author upon reasonable request.

References

- Abraham, A.G., O'Neill, E., 2014. PI3K/Akt-mediated regulation of p53 in cancer. *Biochem. Soc. Trans.* 42 (4), 798–803. <https://doi.org/10.1042/BST20140070>.
- Archana, D., Supriya, C., Girish, B.P., Kishori, B., Reddy, P.S., 2018. Alleviative effect of resveratrol on polyvinyl chloride-induced reproductive toxicity in male Wistar rats. *Food Chem. Toxicol.* 116 (Pt B), 173–181. <https://doi.org/10.1016/j.fct.2018.04.026>.
- Balboa, M.A., Balsinde, J., 2006. Oxidative stress and arachidonic acid mobilization. *Biochim. Biophys. Acta* 1761 (4), 385–391. <https://doi.org/10.1016/j.bbalip.2006.03.014>.
- Banerjee, A., Shelver, W.L., 2021. Micro- and nanoplastic induced cellular toxicity in mammals: a review. *Sci. Total Environ.* 755 (Pt 2), 142518. <https://doi.org/10.1016/j.scitotenv.2020.142518>.
- Besseling, E., Redondo-Hasselerharm, P., Foekema, E.M., Koelmans, A.A., 2018. Quantifying ecological risks of aquatic micro- and nanoplastic. *Crit. Rev. Environ. Sci. Technol.* 49 (1), 32–80. <https://doi.org/10.1080/10643389.2018.1531688>.
- Cave, M., Falkner, K.C., Ray, M., Joshi-Barve, S., Brock, G., Khan, R., McClain, C.J., 2010. Toxicant-associated steatohepatitis in vinyl chloride workers. *Hepatology* 51 (2), 474–481. <https://doi.org/10.1002/hep.23321>.
- Chen, Z., Trotman, L.C., Shaffer, D., Lin, H.K., Dotan, Z.A., Niki, M., Pandolfi, P.P., 2005. Crucial role of p53-dependent cellular senescence in suppression of Pten-deficient tumorigenesis. *Nature* 436 (7051), 725–730. <https://doi.org/10.1038/nature03918>.
- Chen, Y.J., Wu, H., Wu, S.D., Lu, N., Wang, Y.T., Liu, H.N., Shen, X.Z., 2018. Parasutterella, in association with irritable bowel syndrome and intestinal chronic inflammation. *J. Gastroenterol. Hepatol.* 33 (11), 1844–1852. <https://doi.org/10.1111/jgh.14281>.
- Chen, J., Xu, Y., Han, Q., Yao, Y., Xing, H., Teng, X., 2019. Immunosuppression, oxidative stress, and glycometabolism disorder caused by cadmium in common carp (*Cyprinus carpio* L.): application of transcriptome analysis in risk assessment of environmental contaminant cadmium. *J. Hazard. Mater.* 366, 386–394. <https://doi.org/10.1016/j.jhazmat.2018.12.014>.
- Chen, X., Lu, J., Zhao, X., Chen, C., Qiao, D., Wang, H., Yue, X., 2019. Role of C/EBP-beta in methamphetamine-mediated microglial apoptosis. *Front. Cell. Neurosci.* 13, 366. <https://doi.org/10.3389/fncel.2019.00366>.
- Chen, X., Qiu, F., Zhao, X., Lu, J., Tan, X., Xu, J., Wang, H., 2020. Astrocyte-derived Lipocalin-2 is involved in mitochondrion-related neuronal apoptosis induced by methamphetamine. *ACS Chem. Neurosci.* 11 (8), 1102–1116. <https://doi.org/10.1021/acschemneuro.9b00559>.
- Chopyk, D.M., Grakoui, A., 2020. Contribution of the intestinal microbiome and gut barrier to hepatic disorders. *Gastroenterology* 159 (3), 849–863. <https://doi.org/10.1053/j.gastro.2020.04.077>.
- Coleman, R.A., Lee, D.P., 2004. Enzymes of triacylglycerol synthesis and their regulation. *Prog. Lipid Res.* 43 (2), 134–176. [https://doi.org/10.1016/S0163-7827\(03\)00051-1](https://doi.org/10.1016/S0163-7827(03)00051-1).
- Collado, M.C., Isolauri, E., Laitinen, K., Salminen, S., 2008. Distinct composition of gut microbiota during pregnancy in overweight and normal-weight women. *Am. J. Clin. Nutr.* 88 (4), 894–899. <https://doi.org/10.1093/ajcn/88.4.894>.
- Compare, D., Coccoli, P., Rocco, A., Nardone, O.M., De Maria, S., Carteni, M., Nardone, G., 2012. Gut–liver axis: the impact of gut microbiota on non alcoholic fatty liver disease. *Nutr. Metab. Cardiovasc. Dis.* 22 (6), 471–476. <https://doi.org/10.1016/j.numecd.2012.02.007>.
- Deng, Y., Zhang, Y., Lemos, B., Ren, H., 2017. Tissue accumulation of microplastics in mice and biomarker responses suggest widespread health risks of exposure. *Sci. Rep.* 7, 46687. <https://doi.org/10.1038/srep46687>.
- Dessi, C., Okoffo, E.D., O'Brien, J.W., Gallen, M., Samanipour, S., Kaserzon, S., Thomas, K.V., 2021. Plastics contamination of store-bought rice. *J. Hazard. Mater.* 416, 125778. <https://doi.org/10.1016/j.jhazmat.2021.125778>.
- Eerkes-Medrano, D., Thompson, R.C., Aldridge, D.C., 2015. Microplastics in freshwater systems: a review of the emerging threats, identification of knowledge gaps and prioritisation of research needs. *Water Res.* 75, 63–82. <https://doi.org/10.1016/j.watres.2015.02.012>.
- Espinosa, C., Esteban, M.A., Cuesta, A., 2019. Dietary administration of PVC and PE microplastics produces histological damage, oxidative stress and immunoregulation in European sea bass (*Dicentrarchus labrax* L.). *Fish Shellfish Immunol.* 95, 574–583. <https://doi.org/10.1016/j.fsi.2019.10.072>.
- Fuhren, J., Schwalbe, M., Boekhorst, J., Rosch, C., Schols, H.A., Kleerebezem, M., 2021. Dietary calcium phosphate strongly impacts gut microbiome changes elicited by inulin and galacto-oligosaccharides consumption. *Microbiome* 9 (1), 218. <https://doi.org/10.1186/s40168-021-01148-0>.
- Gennaro, V., Ceppi, M., Crosignani, P., Montanaro, F., 2008. Reanalysis of updated mortality among vinyl and polyvinyl chloride workers: confirmation of historical evidence and new findings. *BMC Public Health* 8, 21. <https://doi.org/10.1186/1471-2458-8-21>.
- Gill, S.R., Pop, M., Deboy, R.T., Eckburg, P.B., Turnbaugh, P.J., Samuel, B.S., Nelson, K.E., 2006. Metagenomic analysis of the human distal gut microbiome. *Science* 312 (5778), 1355–1359. <https://doi.org/10.1126/science.1124234>.
- Glenny, E.M., Bulik-Sullivan, E.C., Tang, Q., Bulik, C.M., Carroll, I.M., 2017. Eating disorders and the intestinal microbiota: mechanisms of energy homeostasis and behavioral influence. *Curr. Psychiatry Rep.* 19 (8), 51. <https://doi.org/10.1007/s11920-017-0797-3>.
- Grinkevich, V.V., Nikulenkov, F., Shi, Y., Enge, M., Bao, W.J., Maljukova, A., Selivanova, G., 2009. Ablation of key oncogenic pathways by RITA-reactivated p53 is required for

- efficient apoptosis. *Cancer Cell* 15 (5), 441–453. <https://doi.org/10.1016/j.ccr.2009.03.021>.
- Hagmar, L., Akesson, B., Nielsen, J., Andersson, C., Linden, K., Attewell, R., Moller, T., 1990. Mortality and cancer morbidity in workers exposed to low levels of vinyl chloride monomer at a polyvinyl chloride processing plant. *Am. J. Ind. Med.* 17 (5), 553–565. <https://doi.org/10.1002/ajim.4700170502>.
- Hay, N., 2011. Interplay between FOXO, TOR, and Akt. *Biochim. Biophys. Acta* 1813 (11), 1965–1970. <https://doi.org/10.1016/j.bbamcr.2011.03.013>.
- Herp, S., Durai Raj, A.C., Salvado Silva, M., Woelfel, S., Stecher, B., 2021. The human symbiont *Mucispirillum schaedleri*: causality in health and disease. *Med. Microbiol. Immunol.* 210 (4), 173–179. <https://doi.org/10.1007/s00430-021-00702-9>.
- Iheanacho, S.C., Odo, G.E., 2020. Dietary exposure to polyvinyl chloride microparticles induced oxidative stress and hepatic damage in *Clarias gariepinus* (Burchell, 1822). *Environ. Sci. Pollut. Res. Int.* 27 (17), 21159–21173. <https://doi.org/10.1007/s11356-020-08611-9>.
- Jiang, B., Kauffman, A.E., Li, L., McFee, W., Cai, B., Weinstein, J., Xiao, S., 2020. Health impacts of environmental contamination of micro- and nanoplastics: a review. *Environ. Health Prev. Med.* 25 (1), 29. <https://doi.org/10.1186/s12199-020-00870-9>.
- Jin, Y., Wu, S., Zeng, Z., Fu, Z., 2017. Effects of environmental pollutants on gut microbiota. *Environ. Pollut.* 222, 1–9. <https://doi.org/10.1016/j.envpol.2016.11.045>.
- John, J.A., Smith, F.A., Schwetz, B.A., 1981. Vinyl chloride: inhalation teratology study in mice, rats and rabbits. *Environ. Health Perspect.* 41, 171–177. <https://doi.org/10.1289/ehp.8141171>.
- Jung, S.H., Sim, D.S., Park, M.S., Jo, Q., Kim, Y., 2003. Effects of formalin on haematological and blood chemistry in olive flounder, *Paralichthys olivaceus* (Temminck et Schlegel). *Aquac. Res.* 34 (14), 1269–1275. <https://doi.org/10.1046/j.1365-2109.2003.00936.x>.
- Kataoka, K., 2016. The intestinal microbiota and its role in human health and disease. *J. Med. Invest.* 63 (1–2), 27–37. <https://doi.org/10.2152/jmi.63.27>.
- Kong, C., Gao, R., Yan, X., Huang, L., Qin, H., 2019. Probiotics improve gut microbiota dysbiosis in obese mice fed a high-fat or high-sucrose diet. *Nutrition* 60, 175–184. <https://doi.org/10.1016/j.nut.2018.10.002>.
- Li, S., Tan, H.Y., Wang, N., Zhang, Z.J., Lao, L., Wong, C.W., Feng, Y., 2015. The role of oxidative stress and antioxidants in liver diseases. *Int. J. Mol. Sci.* 16 (11), 26087–26124. <https://doi.org/10.3390/ijms161125942>.
- Li, B., Ding, Y., Cheng, X., Sheng, D., Xu, Z., Rong, Q., Zhang, Y., 2020. Polyethylene microplastics affect the distribution of gut microbiota and inflammation development in mice. *Chemosphere* 244, 125492. <https://doi.org/10.1016/j.chemosphere.2019.125492>.
- Liang, W., Zhou, K., Jian, P., Chang, Z., Zhang, Q., Liu, Y., Zhang, L., 2021. Ginsenosides improve nonalcoholic fatty liver disease via integrated regulation of gut microbiota, inflammation and energy homeostasis. *Front. Pharmacol.* 12, 622841. <https://doi.org/10.3389/fphar.2021.622841>.
- Lu, L., Wan, Z., Luo, T., Fu, Z., Jin, Y., 2018a. Polystyrene microplastics induce gut microbiota dysbiosis and hepatic lipid metabolism disorder in mice. *Sci. Total Environ.* 631–632, 449–458. <https://doi.org/10.1016/j.scitotenv.2018.03.051>.
- Lu, L., Wan, Z.Q., Luo, T., Fu, Z.W., Jin, Y.X., 2018b. Polystyrene microplastics induce gut microbiota dysbiosis and hepatic lipid metabolism disorder in mice. *Sci. Total Environ.* 631–632, 449–458. <https://doi.org/10.1016/j.scitotenv.2018.03.051>.
- Lynch, S.V., Pedersen, O., 2016. The human intestinal microbiome in health and disease. *N. Engl. J. Med.* 375 (24), 2369–2379. <https://doi.org/10.1056/NEJMra1600266>.
- Morrison, D.J., Preston, T., 2016. Formation of short chain fatty acids by the gut microbiota and their impact on human metabolism. *Gut Microbes* 7 (3), 189–200. <https://doi.org/10.1080/19490976.2015.1134082>.
- Nguyen, P., Leray, V., Diez, M., Serisier, S., Le Bloch, J., Siliart, B., Dumon, H., 2008. Liver lipid metabolism. *J. Anim. Physiol. Anim. Nutr. (Berl.)* 92 (3), 272–283. <https://doi.org/10.1111/j.1439-0396.2007.00752.x>.
- Pan, S., Deng, Y., Fu, J., Zhang, Y., Zhang, Z., Ru, X., Qin, X., 2018. Decreased expression of ARHGAP15 promotes the development of colorectal cancer through PTEN/AKT/FOXO1 axis. *Cell Death Dis.* 9 (6), 673. <https://doi.org/10.1038/s41419-018-0707-6>.
- Prata, J.C., da Costa, J.P., Lopes, I., Duarte, A.C., Rocha-Santos, T., 2020. Environmental exposure to microplastics: an overview on possible human health effects. *Sci. Total Environ.* 702, 134455. <https://doi.org/10.1016/j.scitotenv.2019.134455>.
- Qiao, R., Sheng, C., Lu, Y., Zhang, Y., Ren, H., Lemos, B., 2019. Microplastics induce intestinal inflammation, oxidative stress, and disorders of metabolome and microbiome in zebrafish. *Sci. Total Environ.* 662, 246–253. <https://doi.org/10.1016/j.scitotenv.2019.01.245>.
- Rochman, C.M., 2018. Microplastics research—from sink to source. *Science* 360 (6384), 28–29. <https://doi.org/10.1126/science.aar7734>.
- Sadeghi, A., Farokhi, F., Shalazar-Jalali, A., Najafi, G., 2020. Protective effect of vitamin E on sperm quality and in vitro fertilizing potential and testosterone concentration in polyvinyl chloride treated male rats. *Vet. Res. Forum* 11 (3), 257–263. <https://doi.org/10.30466/vrf.2019.91184.2206>.
- Santana, M.F.M., Moreira, F.T., Pereira, C.D.S., Abessa, D.M.S., Turra, A., 2018. Continuous exposure to microplastics does not cause physiological effects in the cultivated mussel *Perna perna*. *Arch. Environ. Contam. Toxicol.* 74 (4), 594–604. <https://doi.org/10.1007/s00244-018-0504-3>.
- Shaidullo, I.F., Sorokina, D.M., Sitdikov, F.G., Hermann, A., Abdulkhakov, S.R., Sitdikova, G.F., 2021. Short chain fatty acids and colon motility in a mouse model of irritable bowel syndrome. *BMC Gastroenterol.* 21 (1), 37. <https://doi.org/10.1186/s12876-021-01613-y>.
- Shen, M., Zhang, Y., Zhu, Y., Song, B., Zeng, G., Hu, D., Ren, X., 2019. Recent advances in toxicological research of nanoplastics in the environment: a review. *Environ. Pollut.* 252 (Pt A), 511–521. <https://doi.org/10.1016/j.envpol.2019.05.102>.
- Taguchi, K., Kensler, T.W., 2020. Nrf2 in liver toxicology. *Arch. Pharm. Res.* 43 (3), 337–349. <https://doi.org/10.1007/s12272-019-01192-3>.
- Wagoner, J.K., 1983. Toxicity of vinyl chloride and poly(vinyl chloride): a critical review. *Environ. Health Perspect.* 52, 61–66. <https://doi.org/10.1289/ehp.835261>.
- Wang, K., Xu, X., Maimaiti, A., Hao, M., Sang, X., Shan, Q., Cao, G., 2021. Gut microbiota disorder caused by diterpenoids extracted from *Euphorbia pekinensis* aggravates intestinal mucosal damage. *Pharmacol. Res. Perspect.* 9 (5), e00765. <https://doi.org/10.1002/prp2.765>.
- Weingarden, A., Vaughn, B.J.G.m., 2017. Intestinal microbiota, fecal microbiota transplantation, and inflammatory bowel disease. 8(3), pp. 238–252. <https://doi.org/10.1080/19490976.2017.1290757>.
- Xia, J., Lu, L., Jin, C., Wang, S., Zhou, J., Ni, Y., Jin, Y., 2018. Effects of short term lead exposure on gut microbiota and hepatic metabolism in adult zebrafish. *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* 209, 1–8. <https://doi.org/10.1016/j.cbpc.2018.03.007>.
- Xu, S., Ma, J., Ji, R., Pan, K., Miao, A.J., 2020. Microplastics in aquatic environments: occurrence, accumulation, and biological effects. *Sci. Total Environ.* 703, 134699. <https://doi.org/10.1016/j.scitotenv.2019.134699>.
- Yao, Y.D., Shen, X.Y., Machado, J., Luo, J.F., Dai, Y., Lio, C.K., Zhou, H., 2019. Nardochinoid B inhibited the activation of RAW264.7 macrophages stimulated by lipopolysaccharide through activating the Nrf2/HO-1 pathway. *Molecules* 24 (13). <https://doi.org/10.3390/molecules24132482>.
- Yu, P., Liu, Z., Wu, D., Chen, M., Lv, W., Zhao, Y., 2018. Accumulation of polystyrene microplastics in juvenile *Eriocheir sinensis* and oxidative stress effects in the liver. *Aquat. Toxicol.* 200, 28–36. <https://doi.org/10.1016/j.aquatox.2018.04.015>.
- Zhang, D.M., Guo, Z.X., Zhao, Y.L., Wang, Q.J., Gao, Y.S., Yu, T., Wang, G.Q., 2019. L-carnitine regulated Nrf2/Keap1 activation in vitro and in vivo and protected oxidized fish oil-induced inflammation response by inhibiting the NF-kappaB signaling pathway in *Rhynchocypris lagowski* Dybowski. *Fish Shellfish Immunol.* 93, 1100–1110. <https://doi.org/10.1016/j.fsi.2019.08.041>.
- Zhang, N., Li, Y.B., He, H.R., Zhang, J.F., Ma, G.S., 2021. You are what you eat: microplastics in the feces of young men living in Beijing. *Sci. Total Environ.* 767, 144345. <https://doi.org/10.1016/j.scitotenv.2020.144345>.
- Zhao, X., Lu, J., Chen, X., Gao, Z., Zhang, C., Chen, C., Wang, H., 2021. Methamphetamine exposure induces neuronal programmed necrosis by activating the receptor-interacting protein kinase 3-related signalling pathway. *FASEB J.* 35 (5), e21561. <https://doi.org/10.1096/fj.202100188R>.